Tavva Dinesh Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.40
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education (TSBIE))	2024	98.30%
Narayana (E.M.) High School	Class X (Board of Secondary Education Andhra Pradesh)	2022	97.66%

SCHOLASTIC ACHIEVEMENTS

- Secured **5th place** out of 751 teams in a national level competition conducted by **EnPower IIT Bombay**.
- Achieved a **maximum rating of 1631** (Expert) in Codeforces (Codeforces Profile)
- Advanced to the Campus Finals of **The Integral Cup** 2025
- Secured **National Top 1% in NSEP** examination 2023.

PROJECTS

Significance-Aware RISC-V Metadata Architecture | GitHub Link

Guide: Prof. Rajesh Kedia | *Computer Architecture*

- Designed a **significance-aware RISC-V architecture** that utilizes **unused upper bits of 64-bit registers** to store **ECC and reliability metadata**.
- Implemented **custom RISC-V ISA extensions** including protected load, significance propagation, fault injection, and adaptive error correction instructions in an **in-house simulator**.
- Integrated a **per-register external protection bit** to enable **selective reliability** without modifying architectural register width.
- Added **SIMD BFloat16 support** and validated ECC recovery through audio and image **fault-injection experiments**.
- Ensured **strict ECC verification** for **control-flow instructions** (branches and jumps) to guarantee safe execution.

xv6 Kernel Development | GitHub Link

- **SC-MLFQ Scheduler:** Built a 4-level System-Call-Aware MLFQ scheduler with dynamic process demotion and global priority boosting to optimize CPU utilization and prevent starvation.
- **Virtual Memory Subsystem:** Implemented demand paging with lazy allocation and a scheduler-aware Clock (second chance) replacement algorithm to intelligently evict memory pages based on process priority.
- **Fault-Tolerant RAID-5 Swap:** Engineered a disk-backed swap mechanism integrated with a software RAID-5 layer, utilizing on-the-fly XOR data reconstruction to ensure data continuity.
- **I/O Dispatch Optimization:** Developed asynchronous, MLFQ-aware disk scheduling supporting FCFS and SSTF policies, reducing seek distances and improving execution time.
- **System Call Interface:** Extended core data structures to support 9 custom system calls, exposing granular process profiling, memory statistics, and system call accounting to user-space.

Effect of Kernel Sizes on Robustness of CNNs | Abstract

Guide: Prof. Anjana A.M.

- Built a **simulation pipeline** to inject **random bit-flip errors** across the 45MB memory footprint of a **ResNet-18 model** to test **hardware fault tolerance**.

- Showed that **increasing convolutional kernel sizes initially improves reliability** against memory errors, but making them **too large degrades performance**.
- **Reduced the memory bandwidth and storage overhead** usually required by **Error Correcting Codes (ECC)** by maximizing built-in fault tolerance through **base architecture tuning**.
- Secured **5th place at ResCon 2025**; **patent filing in progress** for optimizing neural network deployment in fault-prone environments.

Infinite Precision Arithmetic Library | GitHub Link

Guide: Prof. Ashish Mishra | Software Development Fundamentals

- Developed an **object-oriented Java library** (AInteger, AFloat) using **string manipulation** to process numbers that **exceed standard hardware limits**.
- Engineered a **structured build process** with automated execution scripts, packaging the application into a **standalone JAR** to ensure reliable cross-platform execution.

SKILLS

- **Languages:** C, C++, Python, SQL, HTML, CSS, Assembly(RISC-V)
- **Tools & Libraries:** Git, GitHub, Pandas, NumPy

RELEVANT COURSES

- **Computer Science:** Discrete Mathematics, Artificial Intelligence, Software Development Fundamentals, Computer Architecture, Data Structures, Algorithms, Theory of Computation, Operating Systems, Database Management Systems.

- **Mathematics:** Calculus, Differential Equations, Probability and Random Variables, Random Processes, Linear Algebra, Transform Techniques, Statistics.

POSITIONS OF RESPONSIBILITY

- **Introduction to Programming(ID1063):** Assisted as **Teaching Assistant** under Prof. Srijith P.K. and Prof. Saurabh

- **Probability and Random Variables (AI1110):** Assisted as **Teaching Assistant** with course logistics and student doubts under Prof. Anjana A.M.

- **Web Coordinator** at EML(Extra Mural Lectures), IIT Hyderabad